Introduction to Linde and Linde India

Safety, Our First Priority.



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4 Pressure swing adsorption



About The Linde Group

Our One Linde Philosophy

Our Vision

We are committed to fulfilling our vision to be the best performing global gases and engineering company, where our people deliver innovative and sustainable solutions for our customers in a connected world.

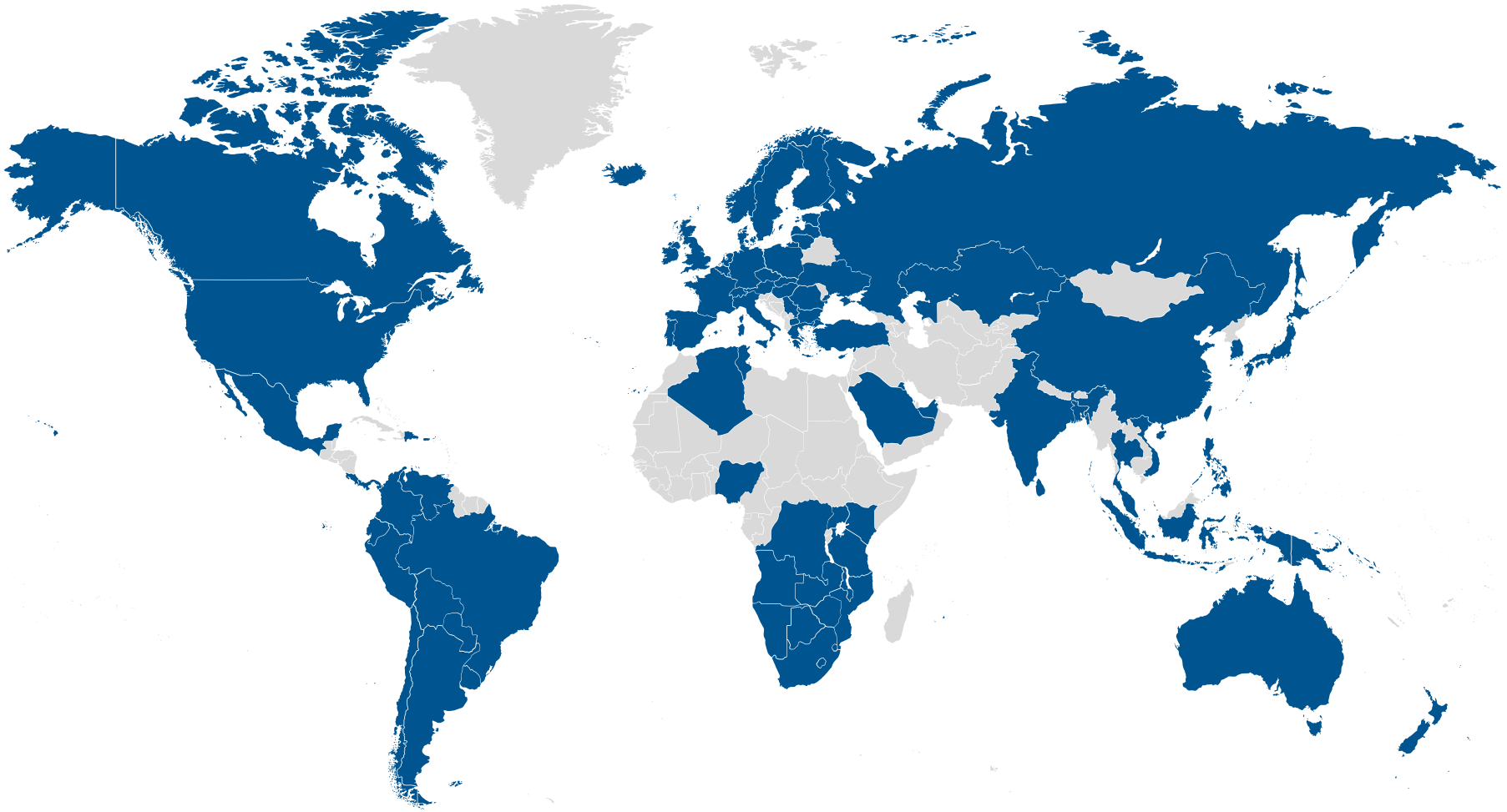
Our Mission

We live our mission of **making our world more productive** every day. Through our high-quality solutions, technologies and services we are making our customers more successful and helping to sustain and protect our planet.

Our Values

- Safety
- Integrity
- Accountability
- Inclusion
- Community

Introduction of Linde

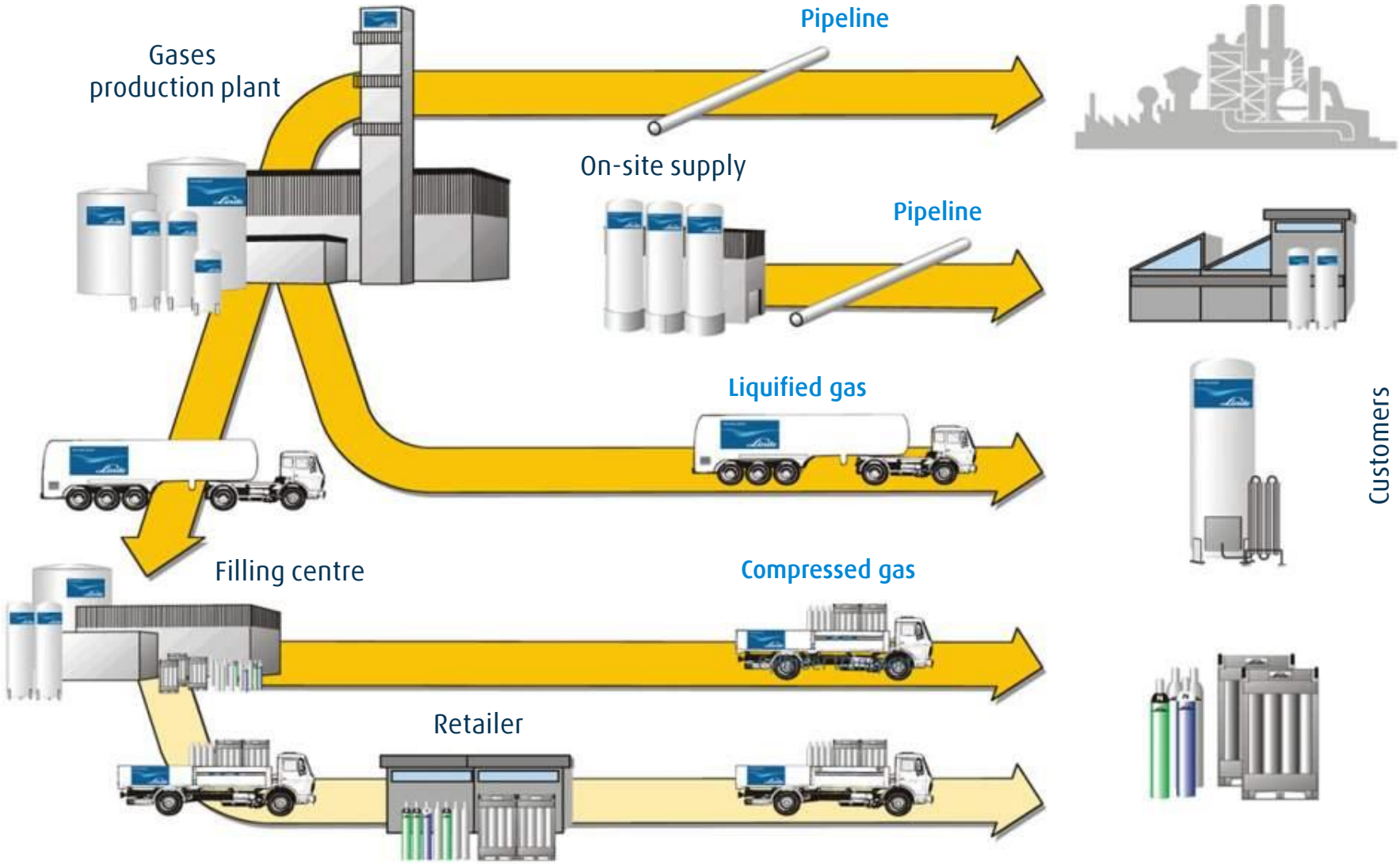


We are the World Largest Company in Gas & Gas Engineering Company with 2023 sales of $32.85 billion.

- Operating in >100 countries
- Largest in India by Capacity & Customer
- Established presence where customers are and where their operations are growing
- Proven critical project execution capability globally

Our Gases Division

Supplying customers through diverse modes meeting different needs.



Our Customer Portfolio

A diverse portfolio... Linde grows with leading companies in key industries.





Linde India

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Introduction of Linde India

- Name of the company : Linde India Ltd.
- Registered Office Address : Oxygen House, P-43, Taratala Road, Kolkata – 700 088.
- Linde India Limited (formerly known as BOC India Limited) having its Head Quarter at Munich, in Germany Linde India Limited is a multinational organization operating in India. Linde Group is the world's largest Gases manufacturing and Engineering company.
- Yearly Turn Over of Linde India Ltd : Over Rs. 3,200 Crores (Approx.)
- Manpower : 700 (approx.)
- List of manufacturing plants in India :

  Major State-of-the Art Tonnage Air Separation plants are located at Jamshedpur, Tarapur, Dahej Taloja, Bellary, Dehradun, Ludhiana and Hyderabad.
- Plant Manufacturing Works is located at Kolkata. Besides above, other plants of Linde India located at Bhiwadi (near Delhi) , Kolkata, Mumbai, Chennai, Bangalore, Hyderabad, Ahmedabad and Visakhapatnam.

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Introduction of Linde India

- **Head Office In Kolkata**
- **ASU Plant at**

  Jamshedpur

  Hyderabad

  Taloja

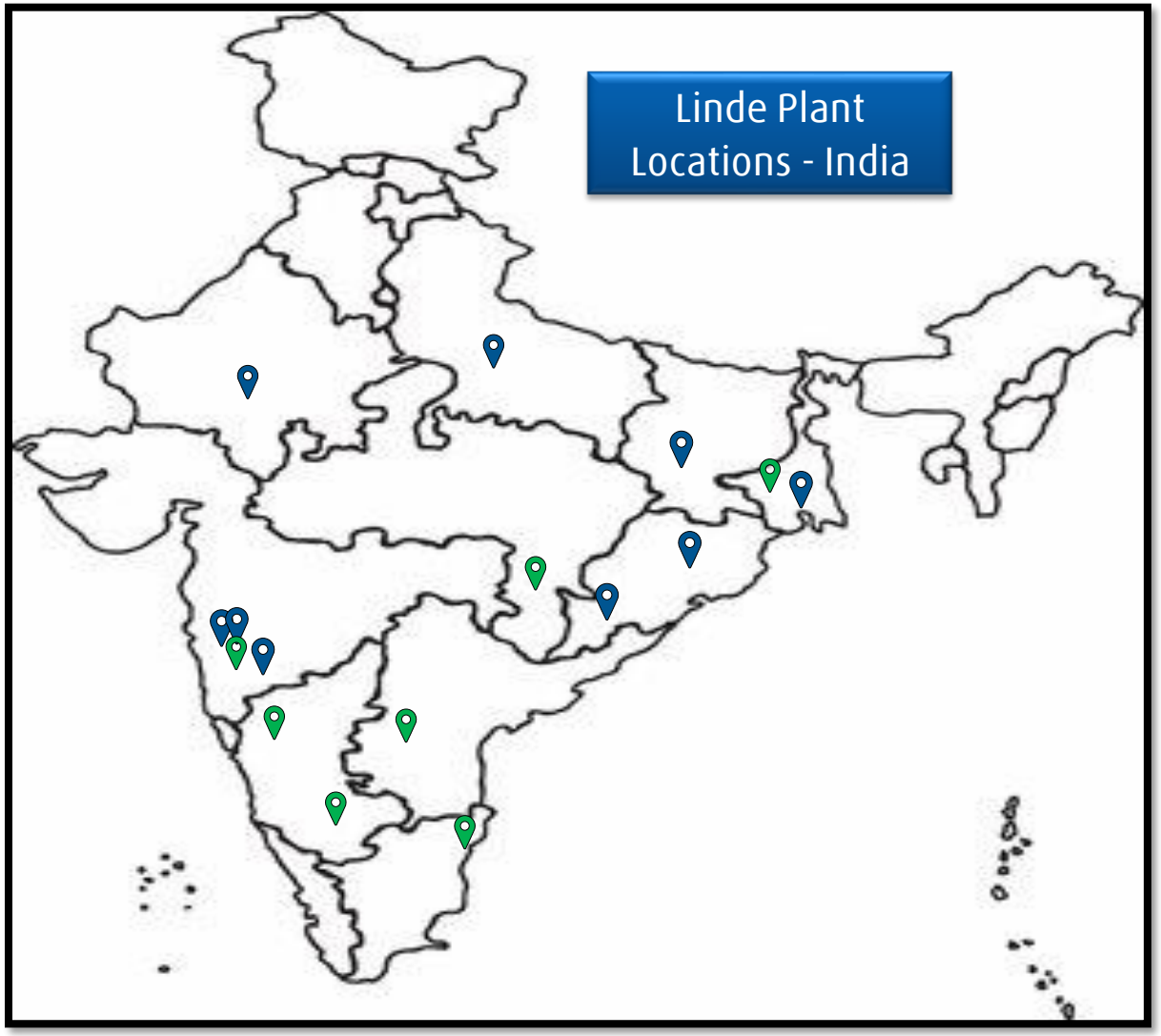
  Bellary

  Kalingonagar

  Derahdoon

  Ludhiana
- **Other Branches**

Ahmedabad	Bangalore
Bhiwadoi	Trichi
Uluberia	Pune
Gurgaon	VIZAG
Mumbai	Cheenai
Siliguri	Rourkela



Introduction of Linde India

- Our company started as THE INDIAN OXYGEN & ACETYLENE COMPANY PRIVATE LIMITED in 1935. The name of the company got changed to INDIAN OXYGEN LIMITED in 1958. The name of the company got changed to IOL LIMITED in 1989. In the year 1995, the name of the company got changed to BOC INDIA LIMITED. The company became LINDE INDIA LIMITED in 2013. In 2018 Globally Linde and Praxiar merged as Linde PLC and India it was happened in 2020.
- Linde India Limited has been in operation for over seven decades and is today a leading gases and engineering company in South Asia, delivering innovative, cost-effective qualitative solutions and services to its customers.
- At Linde India, we are dedicated to provide medical gas products and services, thereby strengthening the country's healthcare infrastructure. In order to enhance patient care, we are committed to partner with healthcare professionals domestically and globally to develop optimal and effective care. The successful model adopted by the healthcare business is the Total Gas Management proposition, where Linde India becomes an integrated part of the hospital by managing gas supply to patients by its own highly skilled and committed team.
- working at the hospital premises. Since inception, we have been working closely with customers for more than Eight decades, addressing the evolving challenges of healthcare.

Introduction of Linde India

- In today's challenging healthcare environment, we offer:
  Medical gases (oxygen, nitrous oxide, air, nitrous oxide/oxygen mixtures, medical helium/oxygen mixtures, medical pulmonary testing gases) etc.

- We also offer solutions and services like: Design, Supply, Installation and Commissioning of centralized medical gas pipeline system. In addition, we also undertake operation and maintenance of medical projects.

- So far, we have installed more than 1000 Centralized Medical Gas Pipeline Systems all over India, on a turnkey basis, which includes prestigious hospitals like:-



Linde Healthcare Overview

Linde Healthcare in India

Strong market leadership with a clear & focused strategy.

Medical gas filling & testing





Medical engineering



Home care



–Market leadership built on strong differentiators & value-added solutions

- Healthcare business presence in 8 of the 11 countries in South & East Asia.
- Strong leadership position in 6 of the markets : Bangladesh, Malaysia, Pakistan, Philippines, Sri Lanka & Thailand.
- In some markets, we also offer turnkey engineering solutions for medical gas supply systems.
- Stringent quality & safety controls during filling & at test-shops.

– Strong growth potential

- Rapid economic development & demographic changes (a growing middle class).
- Aging population (Korea, Singapore).
- Rising cost of healthcare in some markets – potential for home care business.

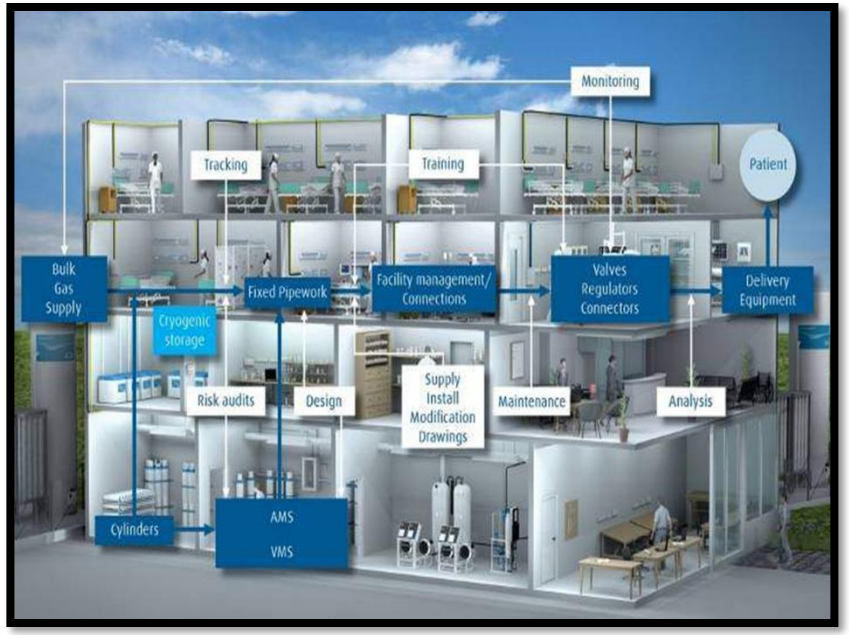
Solutions for Your Daily Medical Gases Needs

Hospital Care - From molecules to infrastructure that delivers it.



One Roof Solution of MGPS+LMO+PSA for Hospital

Complete solution maker with best engineering & Designing for Centralized medical gas pipeline system along with supply of medical oxygen (LMO/CMO/PSA)



MEDICAL GAS PIPELINE SYSTEM



LIQUID MEDICAL OXYGEN



OXYGEN GENERATION PLANT

Medical Gases Packages

Packaging sizes to suit different needs but quality remains the same.

Liquid Supply Solutions, Ranging from 230 litres – 20,000 litres





Gases Supply Solutions, Ranging from 5 litres (W.C.) to 47 litres (W.C.)











Hospital Care Medical Gases

A snapshot on our offer.

Medical Oxygen

O_2

Characteristics

- Colourless
- Odourless
- Non-toxic
- Non-irritant
- Non-flammable but strongly supports combustion



Application

- To maintain tissue oxygenation during surgery
- As part of resuscitation
- To treat hypoxia & hypoxaemia
- To treat COPD, heart attack, etc.

Nitrous Oxide

N_2O

Characteristics

- Colourless
- Slightly sweet taste and odour
- Non-flammable but strongly supports combustion.



Application

- Used as an anaesthetic &analgesic agent:`

Medical Air

Air

Characteristics

- Colourless
- Odourless
- Non-toxic
- Non-irritant
- Non-flammable but supports combustion.



Application

- Mainly used for respiratory therapy
- As carrier gas for narcotic substances in inhalational anaesthesia.

Medical Carbon Dioxide

CO_2

Characteristics

- Colourless
- Odourless
- Non-flammable and does not support combustion
- Forms carbonic acid when in contact with water



Application

- As an insufflation gas for aparoscopy, endoscopy, arthroscopy
- As a cooling agent for cryosurgery

Medical gas Pipeline System













What is Medical Gas Pipe Line System

- Hospitals today need well planned centralised medical Gas Pipeline Systems (MGPS) like other essential internal systems such as Electricity, Water and Air-conditioning.

- Medical Gas Pipeline Systems (MGPS) is an efficient, economic and highly dependable medical life support network that conveys medical gases and vacuum to the patients.

- It's design, installation and commissioning require high degree of skill, technical expertize and experience.

What are the main medical Gases are used in hospitals

- **Oxygen** – Breathing, therapy, ventilators, anesthesia machines
- **Med Air** - Breathing, inhalation, ventilators
- **Vacuum** – Removing secretion
- **Nitrous Oxide (N2O)** - anaesthetization
- **CO2** - Insufflations, Laparoscopic surgery
- **N2/High Pressure Air** – Pneumatic tools (surgery)

Advantages of Centralizes Medical Gas Pipeline System

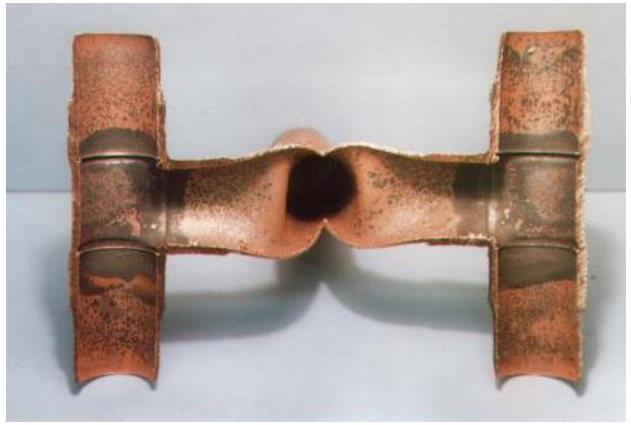
- No distressing sight of oxygen cylinders near the bed.
- Elimination of irksome noise from the movement of cylinders around the hospitals.
- Aseptic, therefore safe.
- Uninterrupted and uncontaminated supply.
- Instant availability of gas on taps.
- Clean, safe and reliable delivery of gases.
- Reduced manual labor.
- Minimal accident hazards due to mishandling of cylinders.
- Continuous flow of gases whenever and wherever required.
- Easy purchase of gases in bulk quantities at favorable terms.
- Economy on purchases of cylinders.
- Fewer breakages.
- Rationalization in ordering, storing and transporting a wide variety and sizes of gas cylinders.
- Saves space.
- Damage to buildings due to handling of cylinders minimized.

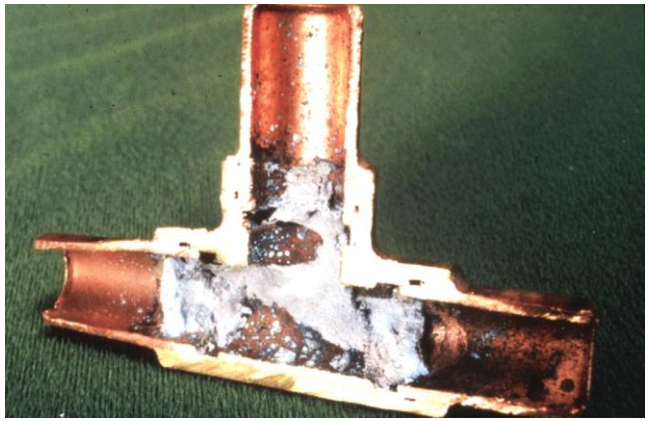
Summary of Source Equipment

Medical Oxygen System	Primary Supply	1) Liquid Oxygen VIE Tank – tap from Existing System or 2) Provide LOX VIE Tank on rental – minimum size of 10k litres
	Secondary Supply	Automatic Changeover Manifold – 2 x 14
Nitrous Oxide System	Primary Supply	Automatic Changeover Manifold – 2 x 2
	Secondary Supply	Reserve Manifold – 3 cylinders
Medical Air & Surgical Air System	Primary Supply	1) Combined Air Plant – FAD : 5320 l/min @ 7 Bar(g) 2) Pressure Reducing Station 7 Bar (g) – 4 Bar (g)
	Secondary Supply	Automatic Changeover Manifold – 2 x 10
Medical Vacuum System	Primary Supply	Medical Vacuum Plant – FAA : 2560 l/min @ 450 mmHg
Anaesthetic Gas Scavenging System	2 sets of pumps	1) Duplex Pumps – serving Level 1 Operating Theatres 2) Duplex Pumps – serving Level 2 Operating Theatres

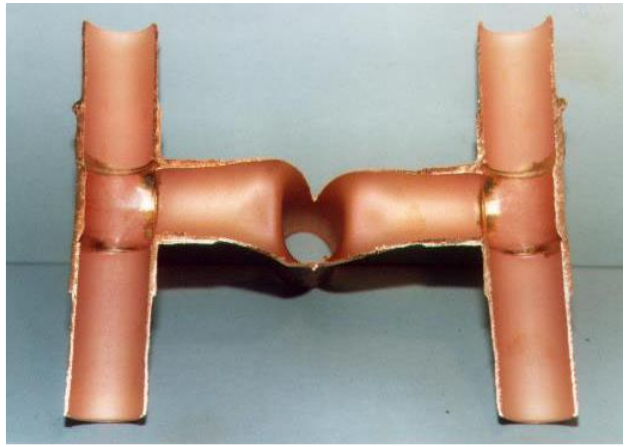
The integrity of the reticulation system is key to delivering quality gases to the patient

The quality of the gases which will be delivered to the patients highly depends on the reticulation system which brings the gas from the source to the end point of use. The installation of the pipeline system must be carried out by qualified and trained personnel.





- Internal condition of pipeline where brazing works are carried out with insufficient nitrogen purging





- Internal condition of pipeline where brazing works are carried out with sufficient nitrogen purging

The reticulation system which is not installed, handled and tested accordingly will contaminate the medical gases that is flowing through

Improper pipe brazing and testing will lead to failure in the final particulate test. Particulates which are present in the Medical Gases will have adverse effects on patients whose lives are dependant on it.



- This test is carried with the actual working gas being filled into the pipeline system
- Gas is being flowed through a particulate filter



- Any contamination found in the pipeline is immediately detected
- As a result, the pipeline system has to be purged or in certain cases the pipeline system has to be replaced. This can lead to extensive cost implications

Detailed testing & commissioning is carried out at every stage of the installation to ensure a safe system is handed over to the hospital

Tests are carried out at various stages throughout the (1) installation phase, (2) upon completion of all the final fixtures and at the (3) final stage when the actual Medical gas is being filled into the pipeline

1





Pressure tests, cross connection , pipeline support and labels are being inspected at this stage

2





Pressure tests, cross connection , performance tests are being carried out at this stage

3





The purity and quality of the Medical Gas Pipeline at the end point of use is checked at this stage

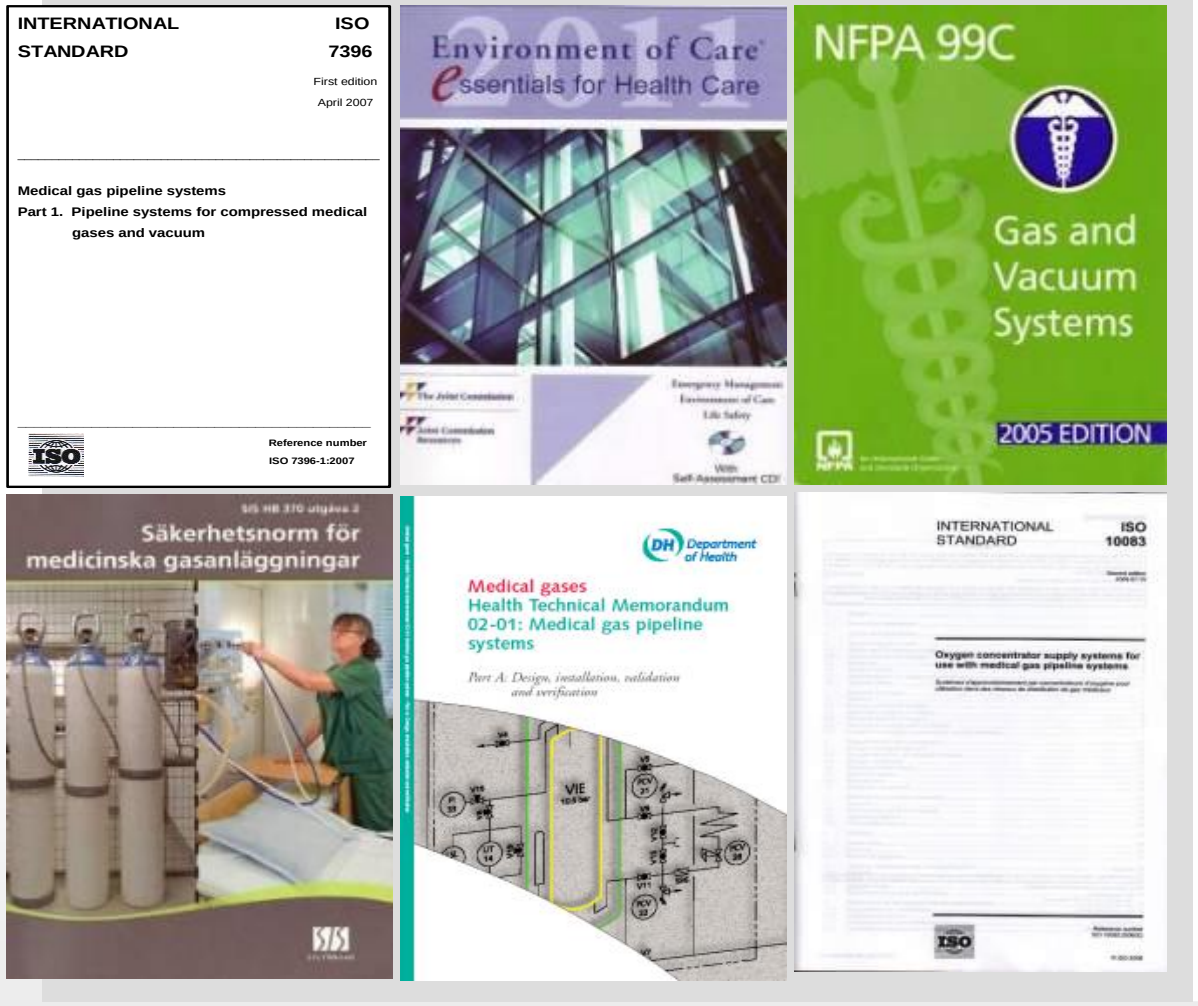
Our Product Assurance
Safety & Quality are our highest priorities.

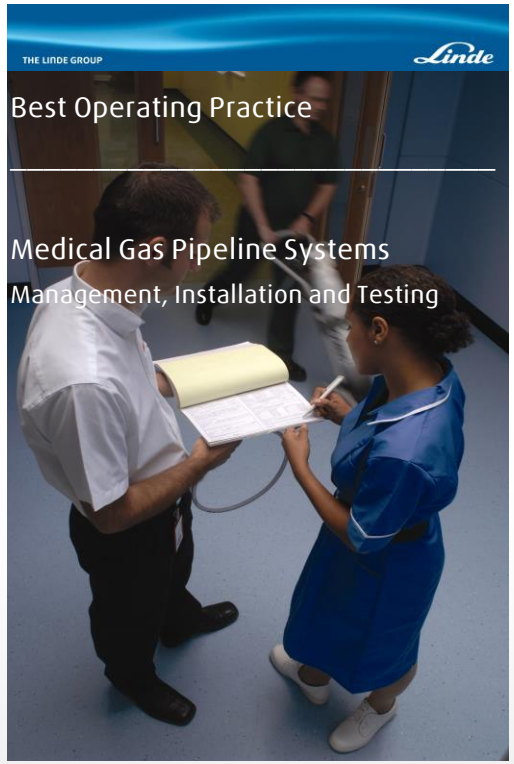
At Linde, safety and quality are our highest priorities to **ensure patients' safety and compliance to regulatory requirements.**

We assure our customers that the patients receive **safe and high quality of medicinal gases consistently,** as expected with any pharmaceutical products, through our **strict implementation of regulatory, industry, and Linde global standards**.



Medical Gas Services complies with Global Standards & Best Operating Practices





In many countries, compliance with the relevant standard is a legal duty.

Installations| Proud Clienteles

Executed 1000+ Medical Gas Pipeline Systems across India:

1. Christian Medical College, Kannighapuram	13. KLE Hospital, Belgaum	25. Ispat General Hospital, SAIL, Rourkela, Odisa.
2. Indira Gandhi Institute of Medical Sciences Patna, Bihar	14. Jorhat Medical College and Hospital, Jorhat, Assam	26. Bokaro General Hospital, Bokaro, Jharkhand
3. NIMS, Hyderabad	15. Jaypee Hospital, Noida	27. J L N Hospital & Research Centre, SAIL-Bhilai, Chhattisgarh
4. Reliance HN Foundation, Mumbai	16. Wockhardt Hospitals Ltd.,Mumbai	28. Fortis Hospital, Kolkata
5. Metro Group of Hospitals – Gurgaon, Haridwar, Jaipur etc.	17. Apollo Hospitals , Motinagar, Delhi	29. Tata Medical Center, Kolkata.
6. Manipal Group of Hospitals – New Delhi, Bengaluru etc..	18. Rajagiri Hospital, Kochi	30. H. N. Hospital, Mumbai.
7. Guahati Medical College and hospital, Guwahati, Assam	19. West Bengal Govt. Med. College and hospitals.	31. Seven Hills, Mumbai
8. KIMS, Bhubaneswar, Odisa	20. ILS Hospital, Agartala, Tripura	32. Deenanath Mangeshkar Hospital and Research Center, Pune
9. Kalpana Chawla Government Medical College, Karnal, Haryana	21. Sir Sunderlal Hospital, BHU, Varanasi, Uttar Pradesh	33. Ruby Hall, Pune
10. Baba Farid Hospital, Bhatinda, Punjab	22. Tata Main Hospital, Jamshedpur, Jharkhand	34. Dr. D. Y. Patil Medical College, Hospital & Research Centre, Pune
11. MGM Hospital & Medical College Aurangabad	23. Jaslok Hospital & Research Center Mumbai	35. Ruby Hall Clinic Pune
12. Virinchi, Hyderabad	24. Antila, Mumbai	36. 6 Civil Hospitals(L & T) – Hissar, Gurgaon, Faridabad, Sonipat, Rohtak & Bhiwani.

Installations| Proud Clienteles

Major Ongoing Projects

- NALCO Hospital - Angul Odisha.
- Imphal Cancer Hospital – Imphal, Manipur.
- Reliance HN Foundation – Mumbai.
- Ruby Hall Clinic – Pune.
- Advanced Veterinary Care Foundation, Tata Trust-Mumbai.
- KIMS – Hiranandani.
- Sai Ashraya Super Speciality Hospital, Bengaluru.
- Manipal Hospital, North – Bengaluru.
- Manipal Hospital, South – Bengaluru.
- Aadhar Health Institute, Hisar, Haryana.
- Sharyana Hospital, Burdwan, West Bengal.

Finished Work





Plant Room





Plant Room





Plant Room





Liquid Medical Oxygen Installation consist of....

Liquid Medical Oxygen consist of..

- Cryogenic storage vessel
- Atmospheric vaporizer
- Operational & engineering directives (O&ED)
- Necessary pipeline



LMO Vessel



Atmospheric Vaporiser



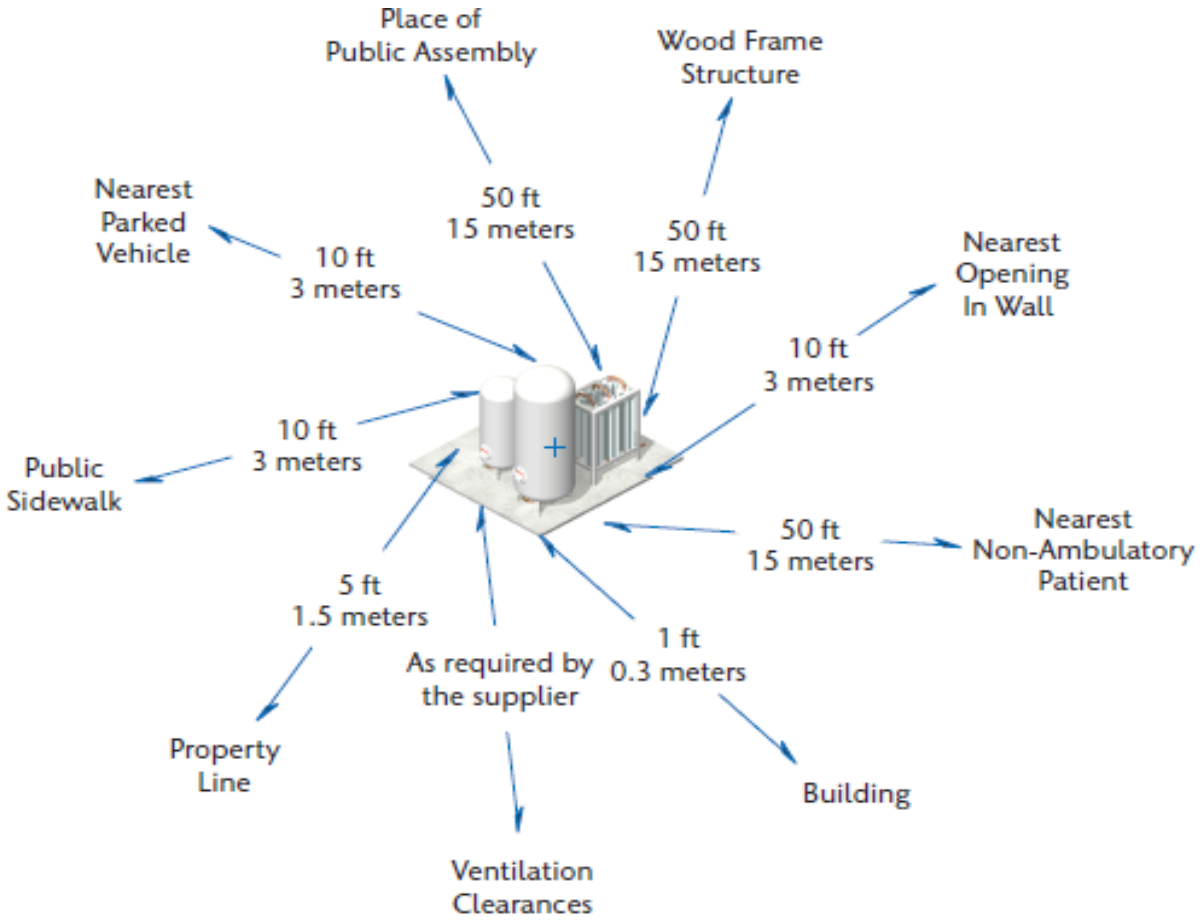


O&ED

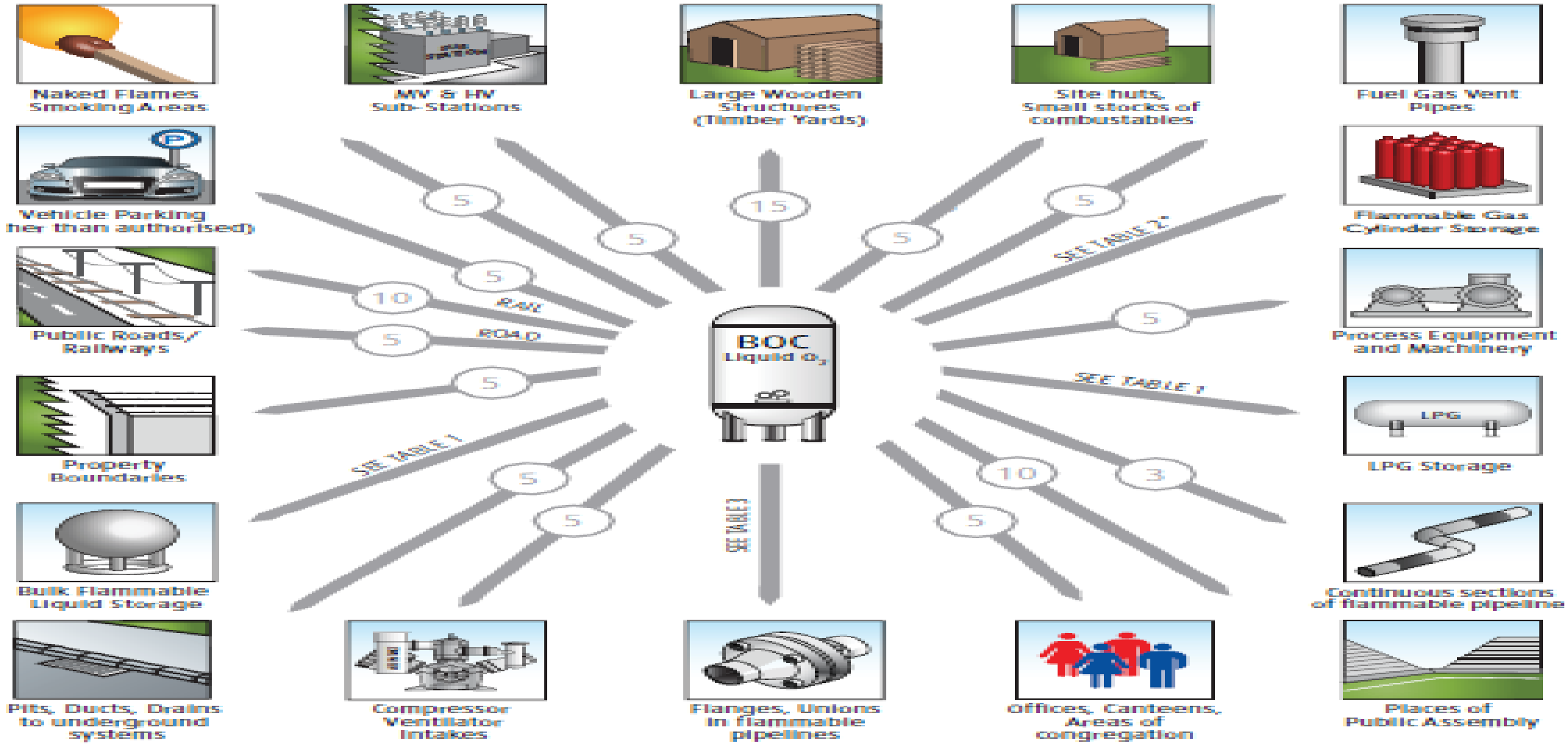
Hospital

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NFPA 50 STANDARD GUIDELINES FOR LMO VESSEL SITE



Minimum Safety distance to be maintained from 1000 Litres LMO VIE to 10000 Litres LMO VIE



Risk Assessment & Aversion

Risk assessment is used to assist

- Development of medical oxygen installation that is Safe
- Has practical Design
- Ensure that a safe supply of oxygen is available for patient use at all times.

Risk Aversion encompasses All aspects of the process -- :

- The initial concept designs
- Installation
- Operation
- Routine assessment of the installation
- Maintenance



Linde's solution (HTM 02- 01)

Medical Operation and Engineering Directive (O&ED) -Key Principles

Telemetry (HTM Standard)

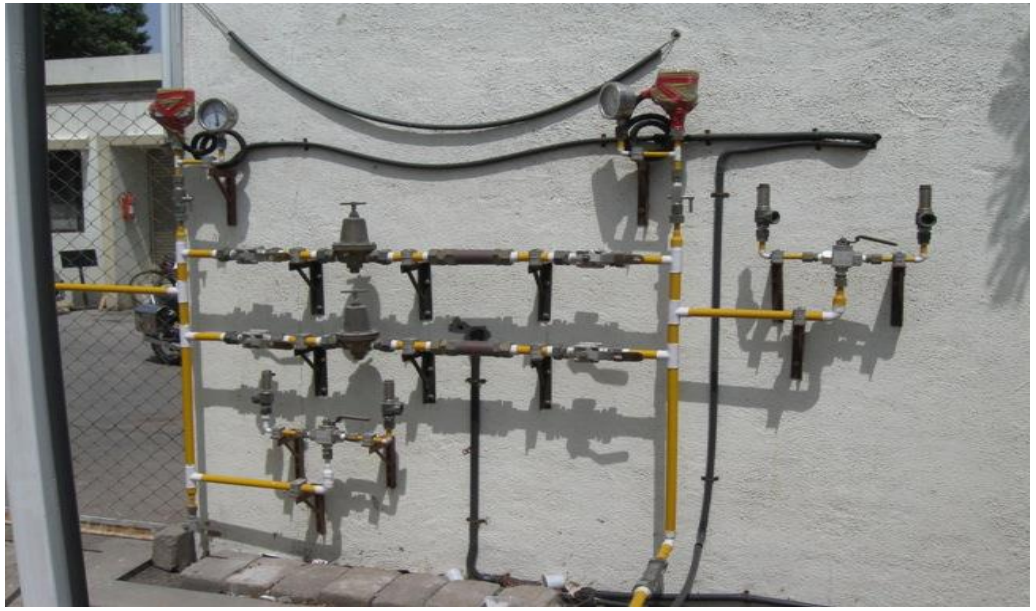
National Scheduling Center

Customer care toll Free no 1800 – 3456789

Exclusive healthcare vertical within organization

Medical Operation and Engineering Directive (O&ED) - **Key Principles**

- Prevent the error due to human element is highly reduced and chances for failure of hospital medical oxygen supply system is eliminated.
- User friendly system.
- Isolation valves in suitable locations for easy maintenance of the system.
- Minimum attendance of operator is required.
- Uninterrupted 24 Hrs oxygen supply to hospital at correct pressure.
- Fool proof, safe, reliable system for gas distribution form source to hospital.
- System designed as per international standard & engineering procedure for hospital gas supply system.



Mandatory alarm signals:

Primary vessel contents are low

Primary vessel pressure is low (Back up supply is in operation).

Hospital pipeline pressure is low

Gives alarm in case of emergency

1. *TWIN PIPELINE SAFETY RELIEF VALVE (SRV) SYSTEM (SRV-1(2nos) and SRV-2 (2nos) as indicated in P&ID):*

• This system has been provided for over pressure protection of oxygen supply pipeline to Hospital. 3-Way diverter valve (DV-1 and DV-2 as indicated in P&ID) ensures that atleast one safety valve is always online and eliminates any chances for over pressurization of Hospital pipe line at any instant of time.

2. *3-WAY GAUGE COCK VALVE (V-5 and V-6 as indicated in P&ID):*

• The 3-Way gauge cock valve is for isolating the line pressure switch with manual vent of locked pressure and is built in 3-Way route and eliminates false low pressure alarm indication.

3. *SEMI AUTO CHANGE OVER SYSTEM FOR CYLINDER MANIFOLD (AC as indicated in P&ID):*

• No operator is required for cylinder bank change over when one bank is exhausted.

• Instantaneous auto change over system ensures no pressure drop in the hospital oxygen gas supply at any time of cylinder bank change over.

Telemetry (HTM Standard)

Telemetry is a mechanism which uses a differential pressure transmitter and an in-built modem to transmit data on the contents of a Liquid Medical Oxygen Vessel, through the internet. This device is mounted on to
the vessel itself and can transmit data. This data is utilized by our National Scheduling Center to monitor the level of stock in the vessel.

- 24 hour world wide data access via the internet.
- Automatic notification of tank replenishment
- Data history helps in forecasting demands

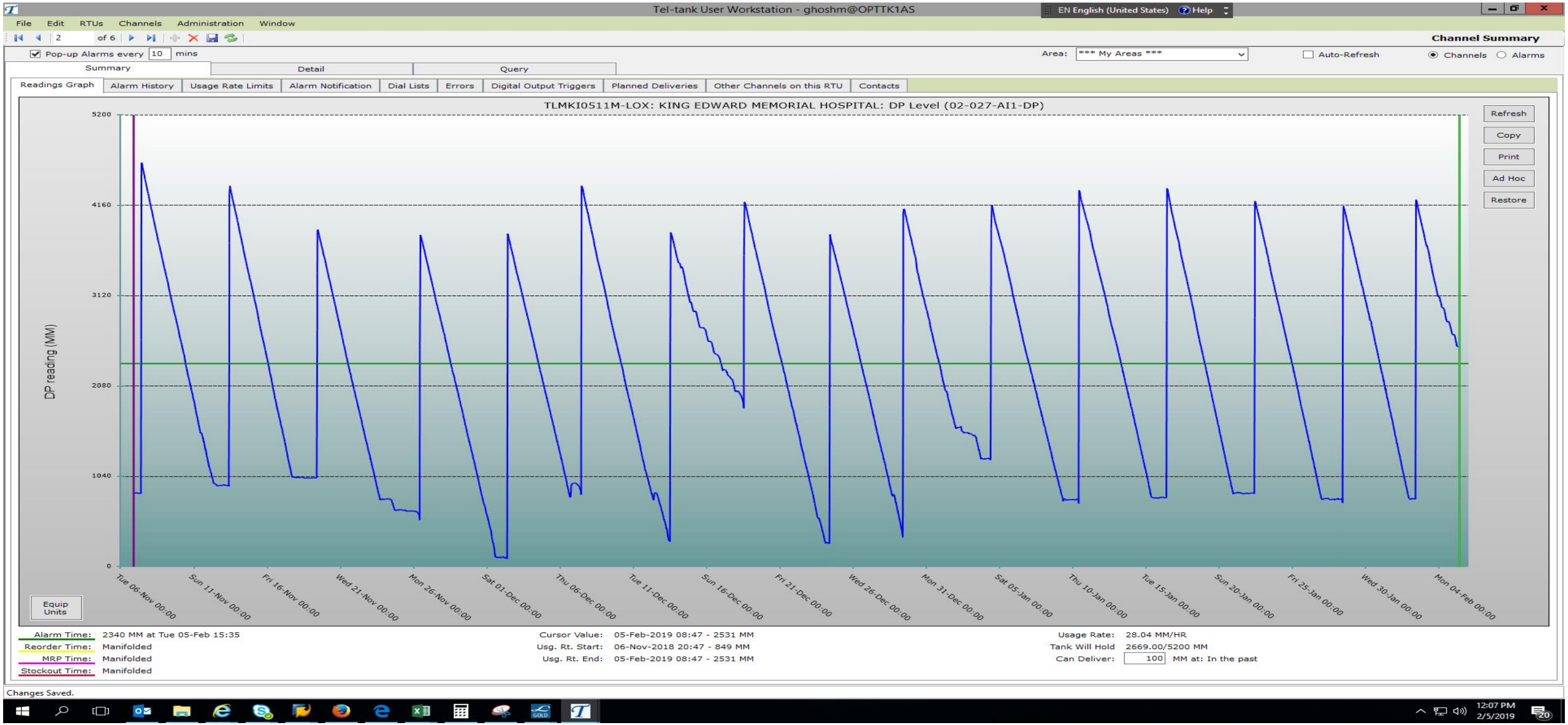
The use of telemetry on the liquid storage system is recommended because it permits both the hospital and the gas supplier to monitor relevant supply conditions continuously, including storage vessel levels and pressure. In addition, it can be used to transmit other operational data from the storage vessel, pipeline and associated equipment for monitoring purposes. (Refer: HTM_02-01_Part_A)

Calculation of primary reserve stock

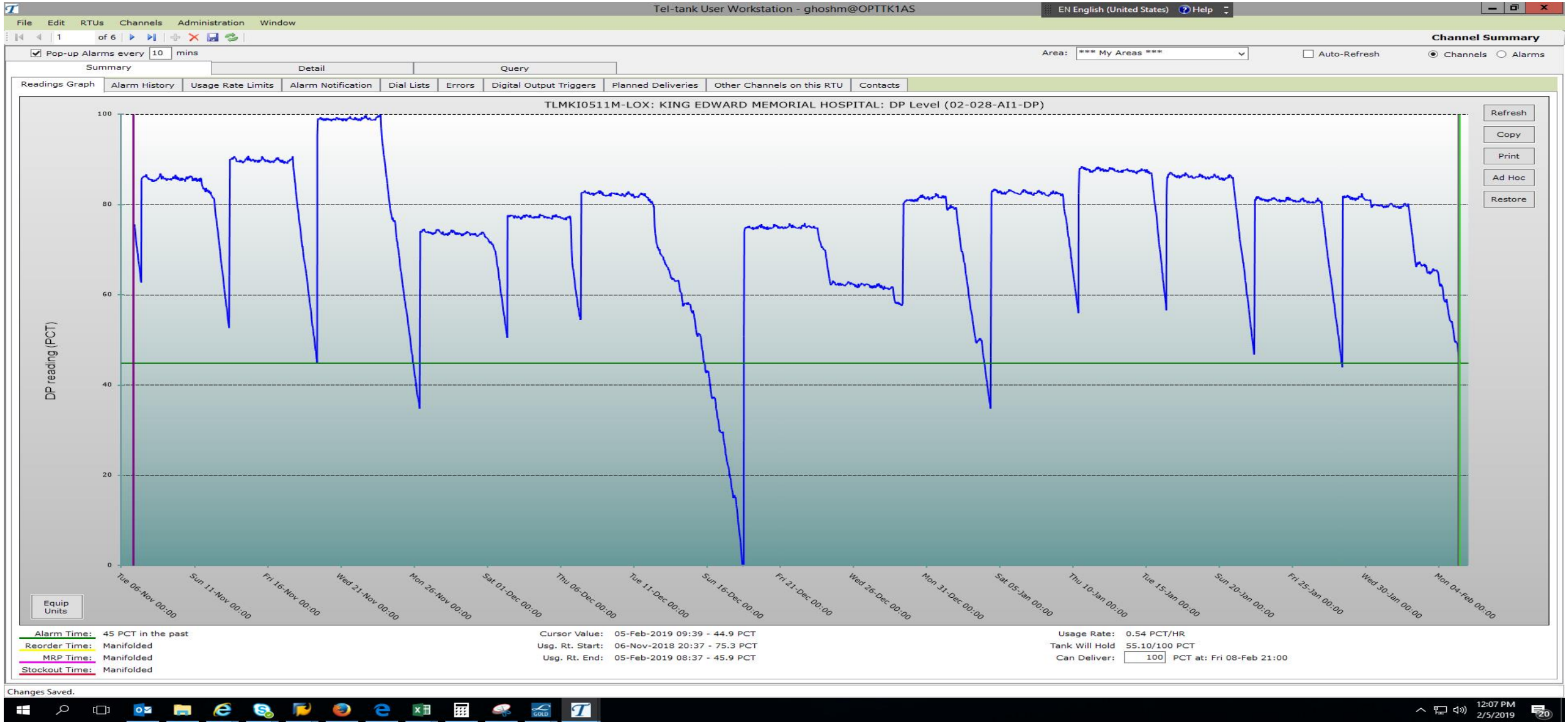
6.160 The table below provides a matrix for the calculation of primary reserve stock based upon distance from gas supplier and fitting of telemetry.

Kilometres from gas supply depot	No telemetry (no of days' stock)	Telemetry fitted (no of days' stock)
Up to 75	5	3
75–150	6	4
150–300	7	5
Over 300	8	6

KEM Data last 6 months (10KL VIE)



KEM Data last 6 months (6KL VIE)



KEM Data last 6 months - Supply

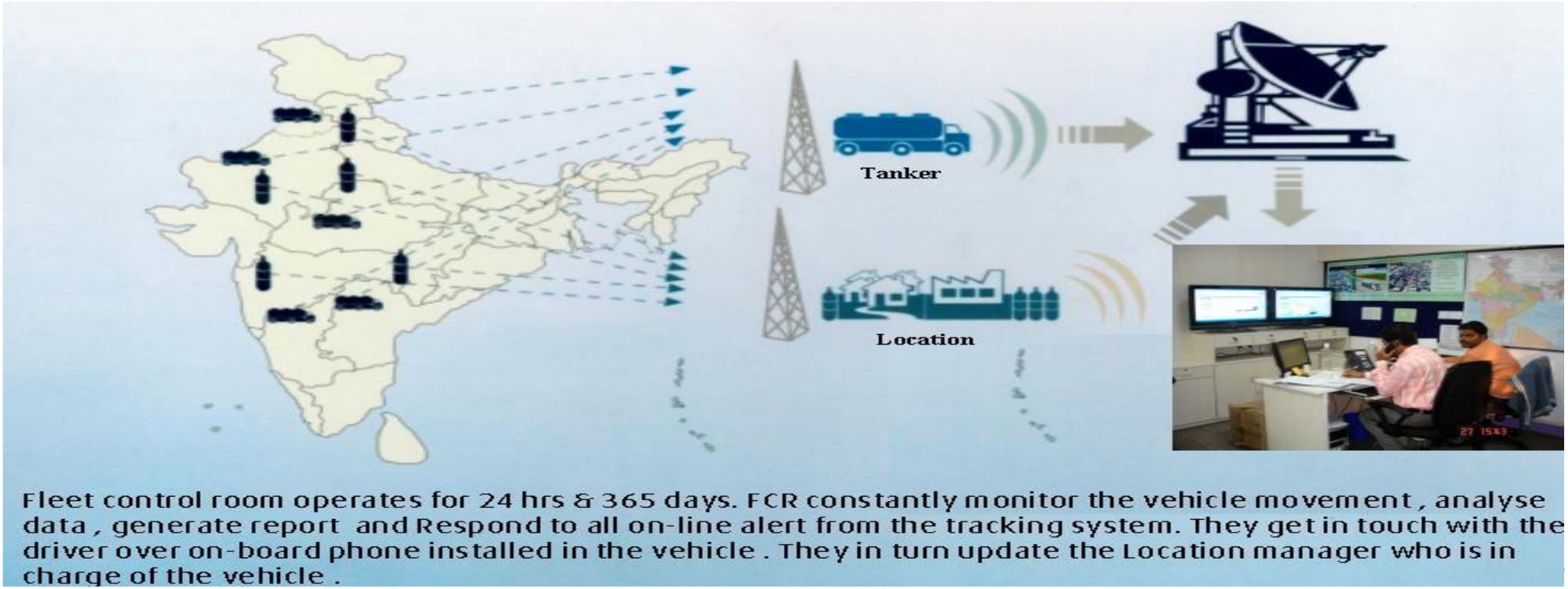
1. Regular & continuous monitoring of Supply (24 X 7)
2. Not a single incident of stock out
3. Order for next supply gets placed automatically
4. No customer intervention

Loc Starting From	Trip Report Number	Actual Finish Date Time	Actual Amount (zm3)
NEW TALOJA: LINDE-TALOJA	8269964	19/11/2018 12:05	7946
NEW TALOJA: LINDE-TALOJA	8264594	13/11/2018 12:12	8193
NEW TALOJA: LINDE-TALOJA	8260229	07/11/2018 10:00	7970
NEW TALOJA: LINDE-TALOJA	8255715	02/11/2018 11:05	8077
NEW TALOJA: LINDE-TALOJA	8252316	29/10/2018 11:15	7977
NEW TALOJA: LINDE-TALOJA	8247970	24/10/2018 11:27	6383
NEW TALOJA: LINDE-TALOJA	8243745	18/10/2018 11:00	8139
NEW TALOJA: LINDE-TALOJA	8237601	12/10/2018 17:40	8254
NEW TALOJA: LINDE-TALOJA	8234156	08/10/2018 13:05	8262
NEW TALOJA: LINDE-TALOJA	8229998	03/10/2018 11:35	8293
NEW TALOJA: LINDE-TALOJA	8224679	27/09/2018 12:00	8385
NEW TALOJA: LINDE-TALOJA	8220548	22/09/2018 12:05	8100
NEW TALOJA: LINDE-TALOJA	8216250	17/09/2018 07:00	8354
NEW TALOJA: LINDE-TALOJA	8211478	11/09/2018 09:00	8331
NEW TALOJA: LINDE-TALOJA	8206803	06/09/2018 14:00	8455
NEW TALOJA: LINDE-TALOJA	8201469	31/08/2018 12:11	7854
NEW TALOJA: LINDE-TALOJA	8196449	25/08/2018 12:20	8085
NEW TALOJA: LINDE-TALOJA	8192217	19/08/2018 14:00	7931
NEW TALOJA: LINDE-TALOJA	8186401	13/08/2018 13:20	8147
NEW TALOJA: LINDE-TALOJA	8180634	06/08/2018 21:35	7954
NEW TALOJA: LINDE-TALOJA	8175660	01/08/2018 06:16	8046

National Scheduling Centre

Only company in India having all India network of well trained service engineers thereby giving excellent service backup.

- All India network of Liquid Medical Oxygen supply.
- Only company with dedicated National Scheduling Center for all Liquid Supplies apart from Customer Care Center.



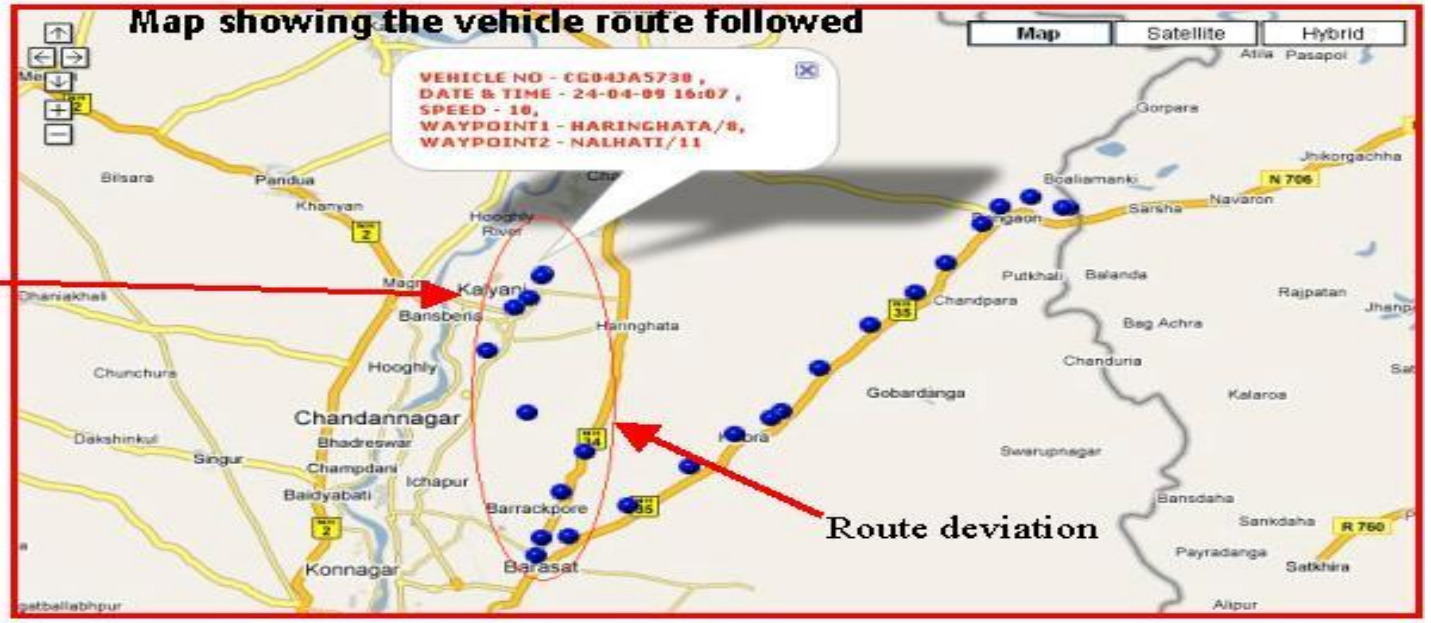
Fleet control room operates for 24 hrs & 365 days. FCR constantly monitor the vehicle movement , analyse data , generate report and Respond to all on-line alert from the tracking system. They get in touch with the driver over on-board phone installed in the vehicle . They in turn update the Location manager who is in charge of the vehicle .

Advantages of 24 X 7 operated National Scheduling Centre

- Faster reaction to manage overnight emergency situations (e.g. truck break down & unable to deliver to customer or sudden surge of customer demand, etc)
- Enables delivery to customers on 24/7 basis due to full time call accessibility to customers
- Related to point no. 2, full time call accessibility to drivers as well
- Provides ample time for Schedulers at late night / early morning shift to prepare good handover notes to daytime shift schedulers
- Overnight changes better managed & communication flow of changes to Depots & day time schedulers on changes will be better since changes would have already been captured in GOLD by then
- Provides ample time for Schedulers at late night / early morning shift to follow up on customer call lists & update level readings for forecasting

On Line Vehicle Tracking - Route Deviation





Trip report of route deviation

DATE & TIME	WAYPOINT1/KM	WAYPOINT2/KM	SPEED
24/04/09 12:15	BANGAON / 2	RANAGHAT / 14	Started -18.4 (25 kmph)
24/04/09 12:25	BANGAON / 1	RANAGHAT / 16	Moving -12 kmph
24/04/09 12:36	BANGAON / 3	GOBARDANGA / 17	Moving -41 kmph
24/04/09 12:45	BANGAON / 8	GOBARDANGA / 12	Moving -21 kmph
24/04/09 12:55	GOBARDANGA / 9	BANGAON / 12	Moving -21 kmph
24/04/09 13:05	GOBARDANGA / 7	HABRA / 14	Moving -19 kmph
24/04/09 13:15	HABRA / 8	GOBARDANGA / 8	Moving -30 kmph
24/04/09 13:26	HABRA / 3	GOBARDANGA / 11	Moving -24 kmph
24/04/09 13:36	HABRA / 2	GOBARDANGA / 12	Moving -4 kmph
24/04/09 13:46	HABRA / 2	HARINGHATA / 15	Moving -3 kmph
24/04/09 13:56	HABRA / 7	BARRACKPORE / 12	Moving -39 kmph
24/04/09 14:06	BARRACKPORE / 7	BARASAT / 8	Moving -41 kmph
24/04/09 14:16	BARASAT / 2	BARRACKPORE / 5	Moving -18 kmph
24/04/09 14:26	BARASAT / 1	BAKRA CROSSING / 6	Stopped
24/04/09 14:36	BARASAT / 1	BARRACKPORE / 5	Started -0.10 (21 kmph)
24/04/09 14:46	**BARRACKPORE / 2**	**BARASAT / 6**	**Moving -36 kmph**
24/04/09 14:56	**BARRACKPORE / 5**	**BARASAT / 11**	**Stopped**
24/04/09 15:16	**NALHATI / 7**	**KANCHRAPARA / 8**	**Moving -38 kmph**
24/04/09 15:36	**NALHATI / 2**	**KANCHRAPARA / 6**	**Moving -40 kmph**
24/04/09 15:47	**NALHATI / 7**	**MAGRA / 9**	**Stopped**
24/04/09 15:57	**NALHATI / 8**	**HARINGHATA / 9**	**Moving -35 kmph**
24/04/09 16:07	**HARINGHATA / 8**	**NALHATI / 11**	**Moving -10 kmph**
24/04/09 16:17	**HARINGHATA / 8**	**MAGRA / 10**	**Stopped**
24/04/09 16:37	**HARINGHATA / 8**	**MAGRA / 10**	**Stopped**
24/04/09 16:47	**HARINGHATA / 8**	**MAGRA / 10**	**Stopped**
24/04/09 16:57	**HARINGHATA / 8**	**MAGRA / 10**	**Stopped**
24/04/09 17:07	**HARINGHATA / 8**	**MAGRA / 10**	**Stopped**
24/04/09 17:17	**HARINGHATA / 8**	**MAGRA / 10**	**Stopped**
24/04/09 17:27	**HARINGHATA / 8**	**MAGRA / 10**	**Stopped**
24/04/09 17:37	**HARINGHATA / 8**	**MAGRA / 10**	**Stopped**
24/04/09 17:47	**HARINGHATA / 8**	**MAGRA / 10**	**Stopped**
24/04/09 17:58	**HARINGHATA / 8**	**MAGRA / 10**	**Stopped**

Vehicle no: CJ04JA5738

Driver : Poonam Mandal

Fact:

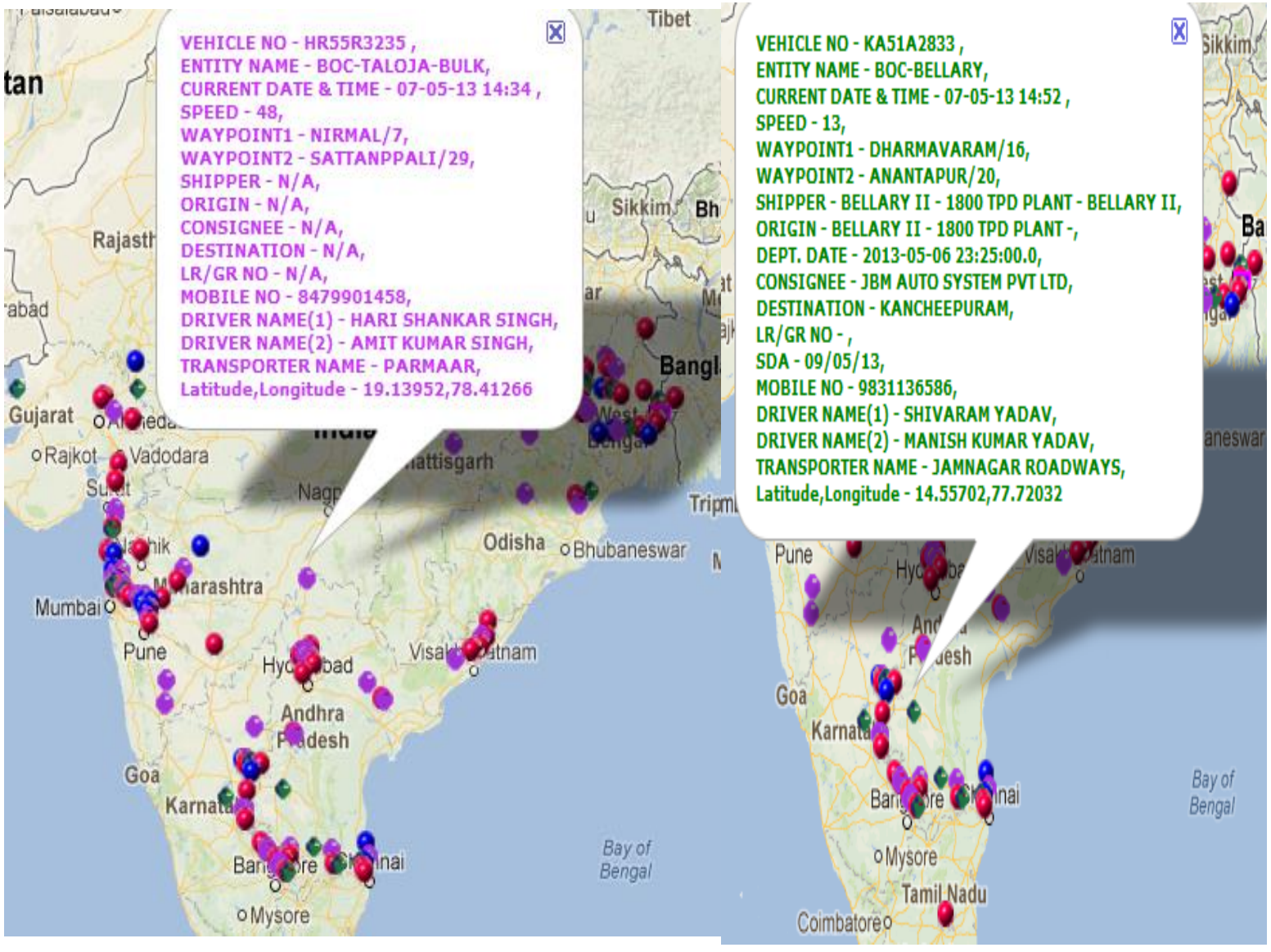
During Return trip from Bangladesh to Jamshedpur , the driver diverted his route from Barasat to Kalyani (West Bengal) in order to visit his house. The same was identified by Fleet control room immediately .

Vehicle Tracking

Polyline Mapping helps in developing Route Map



On Line Vehicle Tracking - Route Deviation



This is done to track the vehicle.....

i- where the vehicle is & where it is going.

ii- weather the vehicle has stopped, if stopped why so!

iii- weather there is an driver issue or a maintenance problem.

iv- to know the road conditions.

v- communicate with the drivers.

DISTRIBUTION TANKERS









Customer care toll Free no 1800 – 3456789

Contactable at any time 24 X 7.

- Reliable information source.
- Dedicated customer care executives in service.
- Complain resolution department on the same number.
- Direct one point of contact with company





Technical Solution

Technical Specification of On-Site O2 Plant



Control Panel with alarm system integrated with PLC

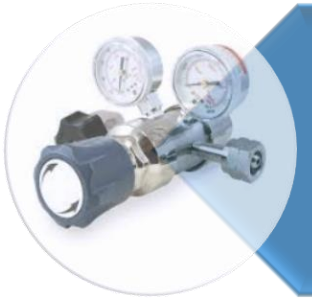


Oxygen Purity: 93%+/-3%



Remote Monitoring

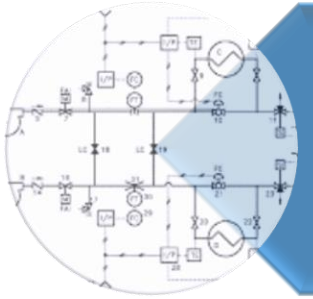




Back Pressure Regulator: If the Hospital's consumption go beyond Plant capacity Plant won't be tripped



The plant will be integrated along with back up system.



Linde will provide plot plan, PFD and P&ID as discussion proceeds with hospital authority

Safety Features



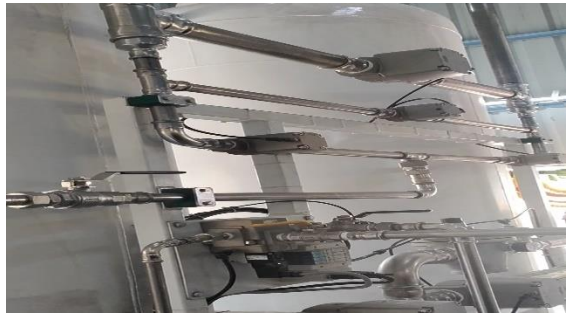
Back Pressure Regulator

- ✓ Helps in maintaining the line pressure
- ✓ Consum>Plant Capa --- Plant will not get tripped & continuously provide maximum output until.



Automatic Hook up and change over system

At any point of time pressure at PSA outlets falls below the set pressure and subsequently the hospital main line pressure also drops, the backup system will come into action automatically and the incremental amount will go from the back up to maintain the required pressure at the hospital pipeline.



Automatic Venting

- ➢ In case of low purity (<90%) Plant will automatically trip.
- ➢ Impure 02 will start venting from 02 receiver
- ➢ Until & unless purity (>/90%) it will be in regeneration mode
- ➢ This feature prevents <90% purity 02 into pipeline.

Oxygen Analyzer & Low Purity Alarm

- ✓ Real time digital Oxygen Purity to monitor the Plant Performance.
- ✓ Audio Visual low purity alarm.



Flowmeter cum totalizer

- ✓ #2 Oxygen Analyzers are installed
- ✓ Location: 1) 02 receiver 2) Main line
- ✓ Give and monitor the real time flow from the Plant.
- ✓ Flowmeter at Hospital line ensure the flow from backup and plant to check the performance of the Plant.



General Scope of Work

S. No	Items (to be provided free of cost, unless specifically mentioned otherwise)	Linde	Customer
1.	Complete Installation and commissioning of Onsite Oxygen plant	✓	
2.	Back-up system of CMO/LMO		✓
3.	Integration of existing back-up system with Production Plant.	✓	
4.	Area for the Oxygen Generation Plant Plant :with free access and free from overhead and underground encumbrances, toxic substances and other contaminants.		✓
5.	Civil layout and foundation data for the plant.	✓	
6.	All civil work.		✓
7.	Power for plant running including Construction power & commissioning power to Linde plant as per specification by Linde. Power supply to the plants (420 V +/-5%, 50 Hz +/-1%) for 15 years contract period with separate cabling from the transformer to plants and further distribution of power line to different equipment of the plant as per voltage requirement will be provided by Hospital.		✓
8.	DG set with automatic & immediate connectivity to the plants in case power outage.		✓
9.	Operational and maintenance Spares	✓	
10.	Complete maintenance.	✓	
11.	Training Program on Onsite oxygen Generation Plant Operation to Hospital. (For operation, manpower will be provided by hospital authority and full training will provided to them by Linde).	✓	
12.	Operation of the Plant		✓

Value engineering points with cost efficiency

- Execution as per Global Standards (HTM / NFPA).
- Dedicated Inhouse qualified experience Design Department.
- Direct Execution -- No Sub-Contracting
- Every project being executed under HTM Certified Project Manager.
- Technical Knowhow more than 100+ Years.
- Most Reliable & Safe MGPS Installation that last for years.
- Global Expertise brings Professionally Managed Expert Execution.
- Ensures The Best Quality for Material.
- Innovation meets Quality & Craftsmanship.
- Highest degree of integrity in Material Handling & Billing.

Only Company who provides

- Full fledge standalone air separation unit(ASU) 200 TPD
- Pump filling facility:
  - Compressed Gas filling facility, 2500 Cylinder per day
  - MBC filling facility, 300 MBCs per day
  - Secondary LMO distribution
  - Full fledge 24 Hrs. service center



Q&A
Thank you for your attention!